

Transverse local stability of the Gaussian propeller optimizer

Run `fa_banach_001`, arXiv:0807.4626, partial result

Source problem

Khot and Naor define an extremal Gaussian partition constant $C(k)$ and prove the equivalent Gaussian-width formula

$$C(k) = \sup \frac{(\mathbb{E} \max_{1 \leq j \leq k} g_j)^2}{\sum_{j=1}^k \mathbb{E} g_j^2},$$

where the supremum is over centered Gaussian k -vectors. They ask whether

$$C(k) \leq C(3) = \frac{9}{8\pi} \quad (k \geq 2).$$

Equivalently, does the global optimizer always use only the three blades of a planar 120° propeller? The question and Gaussian formulation occur on page 20 of [1].

Since $R(3) = 0.3580986219$ it follows that $R(k) < R(3)$ for every integer $k \in [4, 25]$ as well. $\square$

We conjecture that $C(k) \leq C(3)$ for every integer $k \geq 2$. For future reference we end this section with the following alternative characterization of $C(k)$:

Lemma 3.7. We have the following identity:

$$C(k) = \sup \left\{ \frac{(\mathbb{E} [\max_{j \in \{1, \dots, k\}} g_j])^2}{\sum_{j=1}^k \mathbb{E} [g_j^2]} : (g_1, \dots, g_k) \in \mathbb{R}^k \text{ mean zero Gaussian vector} \right\}. \quad (39)$$

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Partial result

We prove strict local maximality against a natural genuinely higher-dimensional class of perturbations: adjoining any fixed finite cloud of small coefficient vectors perpendicular to the propeller plane. The vectors are centered across coordinates, which is the optimal translation normalization for the Gaussian quotient.

Proof intuition

Write M for the maximum of the three propeller coordinates. It dominates a fixed multiple of the planar Gaussian radius, so $\mathbb{P}(M \leq t) = O(t^2)$. A new transverse coordinate of size ε can beat the propeller only on this $O(\varepsilon^2)$ small-ball event, and its payoff there is $O(\varepsilon)$. Thus all the new coordinates improve the expected maximum by only $O(\varepsilon^3)$. Centering the enlarged coefficient family, however, raises its total variance by a strictly positive $O(\varepsilon^2)$ term. The quadratic loss therefore beats the cubic gain for small nonzero ε .

This does not prove the global propeller conjecture: perturbations with longitudinal components, or configurations far from the propeller, remain uncontrolled.

Theorem 1 (Transverse local stability). *Let $E \perp F$ be finite-dimensional real Hilbert spaces. Let $u_1, u_2, u_3 \in E$ satisfy*

$$\|u_i\| = r > 0, \quad \langle u_i, u_j \rangle = -\frac{r^2}{2} \quad (i \neq j),$$

and let $w_1, \dots, w_m \in F$, $m \geq 1$, be fixed and not all zero. Put $k = m + 3$ and

$$\bar{w} = \frac{1}{k} \sum_{j=1}^m w_j.$$

For $\varepsilon \geq 0$, form the centered coefficient family

$$v_i(\varepsilon) = u_i - \varepsilon \bar{w} \quad (1 \leq i \leq 3), \quad v_{3+j}(\varepsilon) = \varepsilon(w_j - \bar{w}) \quad (1 \leq j \leq m).$$

If G is standard Gaussian on $E \oplus F$, define

$$Q(\varepsilon) = \frac{(\mathbb{E} \max_{1 \leq i \leq k} \langle G, v_i(\varepsilon) \rangle)^2}{\sum_{i=1}^k \|v_i(\varepsilon)\|^2}.$$

Then $Q(0) = 9/(8\pi)$, and there is $\varepsilon_0 = \varepsilon_0(u_1, u_2, u_3, w_1, \dots, w_m) > 0$ such that

$$Q(\varepsilon) < Q(0) = \frac{9}{8\pi} \quad (0 < \varepsilon < \varepsilon_0).$$

Proof. The hypotheses on the u_i imply $u_1 + u_2 + u_3 = 0$, and the definition of $\bar{w}$ gives $\sum_{i=1}^k v_i(\varepsilon) = 0$. Write $G = X + Z$, where $X \in E$ and $Z \in F$ are independent standard Gaussians, and set

$$M = \max_{1 \leq i \leq 3} \langle X, u_i \rangle, \quad N = \max_{1 \leq j \leq m} \langle Z, w_j \rangle.$$

Subtracting the same vector $\varepsilon \bar{w}$ from every coefficient adds the common centered Gaussian $-\varepsilon \langle Z, \bar{w} \rangle$ to their maximum. Its expectation is zero, hence

$$\mathbb{E} \max_{1 \leq i \leq k} \langle G, v_i(\varepsilon) \rangle = \mathbb{E} \max\{M, \varepsilon N\}. \quad (1)$$

Put $S = \sum_{i=1}^3 \|u_i\|^2 = 3r^2$. Orthogonality of E and F gives the exact denominator identity

$$\sum_{i=1}^k \|v_i(\varepsilon)\|^2 = S + A\varepsilon^2, \quad A = \sum_{j=1}^m \|w_j\|^2 - \frac{1}{k} \left\| \sum_{j=1}^m w_j \right\|^2. \quad (2)$$

Since $k = m + 3$, Cauchy–Schwarz yields

$$A \geq \left(1 - \frac{m}{k}\right) \sum_{j=1}^m \|w_j\|^2 = \frac{3}{k} \sum_{j=1}^m \|w_j\|^2 > 0.$$

Let $R = \|X\|$. Among three directions separated by 120° , one is within 60° of every nonzero planar vector. Consequently

$$M \geq \frac{r}{2} R \geq 0. \quad (3)$$

As R is the radius of a standard two-dimensional Gaussian, for $t \geq 0$

$$\mathbb{P}(M \leq t) \leq \mathbb{P}(R \leq 2t/r) = 1 - e^{-2t^2/r^2} \leq \frac{2t^2}{r^2}. \quad (4)$$

Let $N_+ = \max\{N, 0\}$ and $\delta(\varepsilon) = \mathbb{E}[\max\{M, \varepsilon N\} - M]$. Conditioning on Z , and using independence together with (4), gives

$$\begin{aligned} 0 \leq \delta(\varepsilon) &= \mathbb{E}(\varepsilon N - M)_+ \\ &\leq \mathbb{E}[\varepsilon N_+ \mathbb{P}(M \leq \varepsilon N_+ \mid Z)] \\ &\leq \frac{2\varepsilon^3}{r^2} \mathbb{E}N_+^3 = B\varepsilon^3, \end{aligned}$$

where $B = 2r^{-2}\mathbb{E}N_+^3 \in (0, \infty)$.

Set $W = \mathbb{E}M > 0$. Equations (1) and (2) imply

$$Q(\varepsilon) \leq \frac{(W + B\varepsilon^3)^2}{S + A\varepsilon^2}.$$

For example, choose

$$\varepsilon_0 = \min \left\{ \frac{WA}{8SB}, \left(\frac{W^2A}{4SB^2} \right)^{1/4} \right\} > 0.$$

If $0 < \varepsilon < \varepsilon_0$, then

$$2SWB\varepsilon + SB^2\varepsilon^4 < W^2A,$$

which is exactly $S(W + B\varepsilon^3)^2 < W^2(S + A\varepsilon^2)$. Hence $Q(\varepsilon) < W^2/S = Q(0)$.

For completeness, the last value can be computed directly. In polar coordinates, $\mathbb{E}R = \sqrt{\pi/2}$, while the angular average of the maximum of three cosines separated by 120° is $3\sqrt{3}/(2\pi)$. Therefore

$$W = r\sqrt{\frac{\pi}{2}} \frac{3\sqrt{3}}{2\pi} = \frac{3\sqrt{3}}{2\sqrt{2}\pi}r, \quad \frac{W^2}{S} = \frac{9}{8\pi}.$$

□

Verification notes

The proof is analytic and has no numerical or computer-assisted dependency. The two delicate points for human review are the centering identity (2) and the cubic gain estimate following (4). Both are displayed explicitly.

A bounded novelty search on 2026-07-22 covered the exact phrases “propeller conjecture”, “Gaussian moments”, “local stability”, and “orthogonal perturbation”, together with arXiv:0807.4626 and its principal later low-dimensional result arXiv:1112.2993. Heilman, Jagannath, and Naor prove the full partition inequality in $\mathbb{R}^3$, while explicitly leaving higher dimensions open [2]. Searches also inspected the related arXiv:1901.03934 and arXiv:2209.11216 abstracts. No exact statement of Theorem 1 was found. Novelty confidence is therefore moderate, not definitive.

Limitations and review recommendation

This is a partial result, not a proof of $C(k) \leq C(3)$. It treats a fixed finite transverse cloud scaled uniformly to zero. It does not treat longitudinal perturbations of the three active vertices, mixed longitudinal–transverse new vertices, or configurations far from the propeller. Status: **candidate partial result, likely valid; human review recommended**, with priority on the $O(\varepsilon^3)$ gain versus $O(\varepsilon^2)$ variance mechanism.

References

- [1] S. Khot and A. Naor, *Approximate kernel clustering*, arXiv:0807.4626, 2008.
- [2] S. Heilman, A. Jagannath, and A. Naor, *Solution of the propeller conjecture in $\mathbb{R}^3$* , arXiv:1112.2993, 2011.